

JESS Thermodynamic Database v8.9

Chemical species in reactions with Titanium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
234TriOHPentDioic-2				
-2	---	39	C(5)H(6)O(7)	178.098
2CoO.TiO2 (s) Dicobalt titanium tetraoxide; 2-Cobalt oxide 1-titanium dioxide				
0	---	1	Co(2)O(4)Ti(1)	229.731
2ThenTriFAcane-1				
-1	---	101	C(8)H(4)F(3)O(2)S(1)	221.174
3CaO.2TiO2 (s) Tricalcium dititanium heptaoxide; 3-Calcium 2-titanium 7-oxide				
0	---	2	Ca(3)O(7)Ti(2)	327.964
4CaO.3TiO2 (s) Tetracalcium trititanium decaoxide; 4-Calcium 3-titanium 10-oxide				
0	---	1	Ca(4)O(10)Ti(3)	463.907
Acac-1 Acetylacetonate ion				
-1	17272-66-1	267	C(5)H(7)O(2)	99.1094
Al (s) Aluminium; Aluminium, cubic				
0	7429-90-5	216	Al(1)	26.9820
Al2TiO5 (s) Dialuminium titanium pentaoxide				
0	---	1	Al(2)O(5)Ti(1)	181.828
Ala-1 Alanine ion; L-Alanine ion; L-2-Aminopropanoate ion				
-1	---	802	C(3)H(6)N(1)O(2)	88.0861

As (s) Arsenic; Arsenic, rhombohedral; Arsenic, grey/gray; Arsenic, alpha				
0	7440-38-2	179	As (1)	74.9220
Asn-1 Asparaginate ion; L-2-Aminobutanedioate 4-amide ion; L-beta-Asparaginate ion; Aspartate beta-amide ion; Altheinate ion; Asparamide ion; Agedoite ion				
-1	---	581	C (4) H (7) N (2) O (3)	131.111
B (s) Boron; Boron, rhombohedral; Boron, beta				
0	7440-42-8	111	B (1)	10.8100
Ba+2 Barium(II) ion				
2	22541-12-4	584	Ba (1)	137.330
Ba (s) Barium; Barium, cubic; Barium, lambda trans.				
0	7440-39-3	64	Ba (1)	137.330
Ba2TiO4 (s) Dibarium titanium tetraoxide				
0	---	1	Ba (2) O (4) Ti (1)	386.525
bAla-1 beta-Alanate ion				
-1	---	498	C (3) H (6) N (1) O (2)	88.0861
BaSrTiO4 (s)				
0	---	1	Ba (1) O (4) Sr (1) Ti (1)	336.815
BaTiO3 (s) Barium titanate				
0	12047-27-7	2	Ba (1) O (3) Ti (1)	233.195
BaTiO4 (s)				
0	---	1	Ba (1) O (4) Ti (1)	249.195
Bipy				

2,2'-Bipyridyl; Bipyridine; 2,2'-Bipyridine; Dipyridyl; 2,2'-Dipyridyl; 2-(2-Pyridyl)pyridine; bipy; bpy; dip; dipy; alpha-alpha'-Bipyridyl				
0	366-18-7	823	C(10)H(8)N(2)	156.187
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
Br2(1) Bromine, liquid				
0	---	285	Br(2)	159.808
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
Ca+2 Calcium(II) ion				
2	14127-61-8	2161	Ca(1)	40.0780
Ca(s) Calcium; Calcium, cubic; Calcium, fcc; Calcium, alpha				
0	7440-70-2	184	Ca(1)	40.0780
CaO.SiO2.TiO2(s) Sphene; Calcium silicon titanium pentaoxide, monoclinic; Titanite				
0	---	2	Ca(1)O(5)Si(1)Ti(1)	196.028
CaO.TiO2(s) Perovskite; Calcium titanium trioxide, pseudo-cubic				
0	---	2	Ca(1)O(3)Ti(1)	135.943
Cat-2 Catecholate ion; 1,2-Dihydroxybenzoate ion; 1,2-Benzenedioate ion; Pyrocatecholate ion; Oxyphenoxide				
-2	19021-48-8	333	C(6)H(4)O(2)	108.097
Cd(s) Cadmium; Cadmium, hexagonal; Cadmium, gamma				
0	7440-43-9	97	Cd(1)	112.410

CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
CdTiO3(s) Cadmium titanium trioxide				
0	---	1	Cd(1)O(3)Ti(1)	208.275
Citric-3 Citrate ion				
-3	126-44-3	1347	C(6)H(5)O(7)	189.102
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060
ClO4-1 Perchlorate ion				
-1	14797-73-0	114	Cl(1)O(4)	99.4506
Co(s) Cobalt; Cobalt, hcp; Cobalt, alpha				
0	7440-48-4	133	Co(1)	58.9330
Cr+3 Chromium(III) ion				
3	16065-83-1	540	Cr(1)	51.9960
Cr+3_Ti+3				
6	---	1	Cr(1)Ti(1)	99.8630
Cupferron-1				
-1	---	34	C(6)H(5)N(2)O(2)	137.118
DTPA-5				

Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C(14)H(18)N(3)O(10)	388.311
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDDA-2 Ethylenediiminodiacetate ion; Ethylenediamine-N,N'-diacetate ion; N,N'-ethylene-diglycinate ion; EDDA ion; N,N'-1,2-ethanediyldisglycinate ion; N,N'-Ethylenediaminediacetate anion				
-2	---	154	C(6)H(10)N(2)O(4)	174.156
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
EtDiAm Ethylenediamine; 1,2-Ethanediamine; 1,2-Diaminoethane; en !				
0	107-15-3	879	C(2)H(8)N(2)	60.0989
F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
F2 (g) Fluorine gas				
0	7782-41-4	259	F(2)	37.9968
Fe+2 Iron(II) ion; Ferrous ion				
2	15438-31-0	1769	Fe(1)	55.8452
Fe+3 Iron(III) ion; Ferric ion				
3	20074-52-6	2512	Fe(1)	55.8452
Fe+3_Ti+3				
6	---	1	Fe(1)Ti(1)	103.712

Fe (s) Iron; Iron, cubic; Iron, bcc				
0	7439-89-6	184	Fe(1)	55.8452
Fe₂TiO₄ (s) Ulvospinel; Diiron titanium tetraoxide, cubic; Diiron titanium tetraoxide				
0	---	1	Fe(2)O(4)Ti(1)	223.555
FeTiO₃ (s) Ilmenite; Iron titanate; Iron titanium trioxide, trigonal; Iron titanium trioxide				
0	12022-71-8	2	Fe(1)O(3)Ti(1)	151.710
Gluconic-1 Gluconate ion				
-1	---	133	C(6)H(11)O(7)	195.149
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_234THB-4				
-3	---	2	C(7)H(3)O(5)	167.098
H+1_Aniline				
1	---	2	C(6)H(8)N(1)	94.1362
H+1_Cat-2 o-Hydroxyphenoxide				
-1	---	26	C(6)H(5)O(2)	109.105
H+1_CDTA-4				
-3	---	39	C(14)H(19)N(2)O(8)	343.313
H+1_Citric-3				
-2	---	93	C(6)H(6)O(7)	190.109

H+1_DTPA-5				
-4	---	44	C(14)H(19)N(3)O(10)	389.319
H+1_EDTA-4				
-3	---	70	C(10)H(13)N(2)O(8)	289.222
H+1_F-1				
0	---	142	F(1)H(1)	20.0063
H+1_Gly-1 Glycine; Aminoacetic acid; Aminoethanoic acid; 2-Aminoethanoic acid				
0	---	82	C(2)H(5)N(1)O(2)	75.0672
H+1_NTA-3				
-2	---	33	C(6)H(7)N(1)O(6)	189.125
H+1_PAR-2				
-1	---	46	C(11)H(8)N(3)O(2)	214.203
H+1_PO4-3 Hydrogenphosphate ion				
-2	---	317	H(1)O(4)P(1)	95.9795
H+1_Pyrogallol-3				
-2	---	12	C(6)H(4)O(3)	124.096
H+1_Salicylic-2				
-1	---	111	C(7)H(5)O(3)	137.115
H+1_SO4-2 Hydrogensulfate ion				
-1	---	83	H(1)O(4)S(1)	97.0655
H+1_SulfSal-3				
-2	---	54	C(7)H(4)O(6)S(1)	216.165
H+1_Tartaric-2				

-1	---	39	C (4) H (5) O (6)	149.080
H+1_Tiron-4				
-3	---	29	C (6) H (3) O (8) S (2)	267.205
H+1 (2)_234THB-4				
-2	---	12	C (7) H (4) O (5)	168.106
H+1 (2)_24DHB-3				
-1	---	27	C (7) H (5) O (4)	153.114
H+1 (2)_CDTA-4				
-2	---	8	C (14) H (20) N (2) O (8)	344.321
H+1 (2)_Chromotropic-4				
-2	---	9	C (10) H (6) O (8) S (2)	318.273
H+1 (2)_Citric-3				
-1	---	39	C (6) H (7) O (7)	191.117
H+1 (2)_EDTA-4				
-2	---	37	C (10) H (14) N (2) O (8)	290.230
H+1 (2)_Salicylic-2				
0	---	15	C (7) H (6) O (3)	138.123
H+1 (2)_SiH2O4-2 Silicic acid				
0	1343-98-2	184	H (4) O (4) Si (1)	96.1149
H+1 (2)_SulfSal-3				
-1	---	10	C (7) H (5) O (6) S (1)	217.173
H2 (g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H (2)	2.01588

H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
H2O2 Hydrogen peroxide, aqueous				
0	---	27	H(2)O(2)	34.0147
HO2-1 Hydrogen peroxide anion; Peroxide ion				
-1	---	12	H(1)O(2)	33.0067
HTiO3-1				
-1	---	1	H(1)O(3)Ti(1)	96.8731
I-1 Iodide ion				
-1	20461-54-5	894	I(1)	126.900
I2(s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I(2)	253.800
Lactic-1 Lactate ion				
-1	---	643	C(3)H(5)O(3)	89.0709
Leu-1 Leucinate ion; L-Leucinate ion; 2-amino-4-methylvalerate ion; alpha-aminoisocaproate ion; 2-amino-4-methylpentanoate ion; L-2-amino-4-methylpentanoate ion				
-1	17332-93-3	511	C(6)H(12)N(1)O(2)	130.167
Li(s) Lithium; Lithium, cubic				
0	7439-93-2	62	Li(1)	6.94100
Li2TiO3(s) Dilithium titanium trioxide				
0	---	1	Li(2)O(3)Ti(1)	109.747

Malic-2 Malate ion				
-2	149-61-1	682	C(4)H(4)O(5)	132.073
Malonic-2 Malonate ion				
-2	156-80-9	417	C(3)H(2)O(4)	102.047
MeMalon-2				
-2	---	55	C(4)H(4)O(4)	116.073
Mg+2 Magnesium(II) ion				
2	22537-22-0	2035	Mg(1)	24.3050
Mg (s) Magnesium; Magnesium, hexagonal				
0	7439-95-4	154	Mg(1)	24.3050
Mg₂TiO₄ (s) Magnesium orthotitanate; Dimagnesium titanium tetraoxide				
0	---	2	Mg(2)O(4)Ti(1)	160.475
MgO.2TiO₂ (s) Magnesium dititanium pentoxide				
0	---	2	Mg(1)O(5)Ti(2)	200.036
MgTiO₃ (geik, s) Magnesium metatitanate; Magnesium titanium tetraoxide, rhombohedral; Geikielite; Magnesium titanium tetraoxide; Magnesium titanate				
0	1312-99-8	2	Mg(1)O(3)Ti(1)	120.170
MgTiO₃ (meta, s) Metatitanate				
0	---	1	Mg(1)O(3)Ti(1)	120.170
Mn (s) Manganese; Manganese, alpha; Manganese, complex bcc				
0	7439-96-5	72	Mn(1)	54.9380

Mn₂TiO₄ (s) Manganese orthotitanate; Dimanganese titanium tetraoxide				
0	---	1	Mn(2)O(4)Ti(1)	221.741
MnTiO₃ (s) Manganese metatitanate; Manganese titanium trioxide, rhombohedral; Pyrophanite; Manganese titanium trioxide				
0	---	1	Mn(1)O(3)Ti(1)	150.803
N₂ (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N(2)	28.0134
Na (s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na(1)	22.9900
Na₂Ti₂O₅ (s) Disodium dititanium pentaoxide				
0	---	1	Na(2)O(5)Ti(2)	221.711
Na₂Ti₃O₇ (s) Disodium trititanium heptaoxide				
0	---	1	Na(2)O(7)Ti(3)	301.577
Na₂TiO₃ (s) Sodium metatitanate; Disodium titanium trioxide				
0	---	1	Na(2)O(3)Ti(1)	141.845
Ni (s) Nickel; Nickel, fcc				
0	7440-02-0	148	Ni(1)	58.6930
NiTiO₃ (s) Nickel titanium trioxide				
0	---	1	Ni(1)O(3)Ti(1)	154.558
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				
-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117

O2-2				
-2	---	6	O (2)	31.9988
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O (2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H (1) O (1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C (2) O (4)	88.0196
Oxine-1 8-Hydroxyquinolate ion; Oxine ion				
-1	16582-16-4	162	C (9) H (6) N (1) O (1)	144.153
Pb+2 Lead(II) ion				
2	14280-50-3	2437	Pb (1)	207.200
Pb+2_TiO+2_Oxalic-2 (2)_(s)				
0	---	2	C (4) O (9) Pb (1) Ti (1)	447.106
Pb (s) Lead; Lead, cubic				
0	7439-92-1	127	Pb (1)	207.200
PbTiO3 (s) Lead titanium trioxide				
0	---	1	O (3) Pb (1) Ti (1)	303.065
Phenanth 1,10-Phenanthroline; phen; Phenanthroline; Orthophenathroline (sic)				
0	66-71-7	406	C (12) H (8) N (2)	180.209
Picolinic-1 Picolinate ion				

-1	3198-27-4	208	C(6)H(4)N(1)O(2)	122.103
PO4-3 Phosphate ion; Phosphate ion				
-3	14265-44-2	1594	O(4)P(1)	94.9716
Pro-1 Prolinate ion; L-Prolinate ion; 2-pyrrolidinecarboxylate ion; L-pyrrolidine-2-carboxylate ion				
-1	17781-82-7	494	C(5)H(8)N(1)O(2)	114.124
S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600
Salicylaldoxime-2				
-2	---	35	C(7)H(5)N(1)O(2)	135.122
Salicylic-2 Salicylate ion				
-2	16887-56-2	558	C(7)H(4)O(3)	136.107
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C(1)N(1)S(1)	58.0777
Ser-1 Serinate ion; L-Serinate ion; 2-amino-3-hydroxypropionate ion; beta-hydroxyalanate ion; 2-amino-3-hydroxypropanoate ion; alpha-amino-beta-hydroxypropionate ion; L-2-amino-3-hydroxypropanoate ion				
-1	17807-54-4	708	C(3)H(6)N(1)O(3)	104.086
Si(s) Silicon; Silicon, cubic				
0	7440-21-3	246	Si(1)	28.0855
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576

Sr+2 Strontium(II) ion				
2	22537-39-9	693	Sr(1)	87.6200
Sr(s) Strontium; Strontium, cubic; Strontium, alpha				
0	7440-24-6	55	Sr(1)	87.6200
Sr2TiO4(s) Distrontium titanium tetraoxide				
0	---	1	O(4)Sr(2)Ti(1)	287.105
Sr4Ti3O10(s) Tetrastrontium trititanium decaoxide				
0	---	1	O(10)Sr(4)Ti(3)	654.075
SrTiO3(s) Strontium titanate; Strontium titanium oxide				
0	---	3	O(3)Sr(1)Ti(1)	183.485
SrTiO4(s)				
0	---	1	O(4)Sr(1)Ti(1)	199.485
SulfSal-3 5-Sulfosalicylate ion				
-3	---	238	C(7)H(3)O(6)S(1)	215.157
Tartaric-2 Tartrate ion				
-2	---	393	C(4)H(4)O(6)	148.072
Ti+2 Titanium(II) ion				
2	15969-58-1	11	Ti(1)	47.8670
Ti+2_Br-1(2)_ (s) Titanium(II) bromide; Titanium dibromide				
0	---	1	Br(2)Ti(1)	207.675
Ti+2_Cl-1				

1	---	1	Cl (1) Ti (1)	83.3200
Ti+2_Cl-1(2)_ (s) Titanium(II) chloride; Titanium dichloride				
0	---	1	Cl (2) Ti (1)	118.773
Ti+2_F-1(2)_ (s)				
0	---	1	F (2) Ti (1)	85.8638
Ti+2_I-1(2)_ (s) Titanium(II) iodide; Titanium diiodide				
0	---	1	I (2) Ti (1)	301.667
Ti+2_S-2_ (s) Titanium(II) sulfide; Titanium monosulfide				
0	---	1	S (1) Ti (1)	79.9270
Ti+2(3)_AsO4-3(2)_ (s) Titanium(II) arsenate; Trititanium diarsenate				
0	---	1	As (2) O (8) Ti (3)	421.440
Ti+3 Titanium(III) ion				
3	22541-75-9	106	Ti (1)	47.8670
Ti+3_Acac-1				
2	---	1	C (5) H (7) O (2) Ti (1)	146.976
Ti+3_Acac-1(2)				
1	---	1	C (10) H (14) O (4) Ti (1)	246.086
Ti+3_Acac-1(3)				
0	---	1	C (15) H (21) O (6) Ti (1)	345.195
Ti+3_Ala-1				
2	---	1	C (3) H (6) N (1) O (2) Ti (1)	135.953
Ti+3_Asn-1				
2	---	1	C (4) H (7) N (2) O (3) Ti (1)	178.978

Ti+3_bAla-1				
2	---	2	C(3)H(6)N(1)O(2)Ti(1)	135.953
Ti+3_bAla-1(2)				
1	---	2	C(6)H(12)N(2)O(4)Ti(1)	224.039
Ti+3_Bipy				
3	---	1	C(10)H(8)N(2)Ti(1)	204.054
Ti+3_Bipy(2)				
3	---	1	C(20)H(16)N(4)Ti(1)	360.241
Ti+3_Bipy(3)				
3	---	1	C(30)H(24)N(6)Ti(1)	516.428
Ti+3_Br-1(3)_(s) Titanium(III) bromide; Titanium tribromide				
0	---	2	Br(3)Ti(1)	287.579
Ti+3_Cl-1				
2	---	1	Cl(1)Ti(1)	83.3200
Ti+3_Cl-1(2)				
1	---	2	Cl(2)Ti(1)	118.773
Ti+3_Cl-1(3)_(s) Titanium(III) chloride; Titanium trichloride				
0	---	2	Cl(3)Ti(1)	154.226
Ti+3_EDDA-2				
1	---	2	C(6)H(10)N(2)O(4)Ti(1)	222.023
Ti+3_EDDA-2(2)				
-1	---	1	C(12)H(20)N(4)O(8)Ti(1)	396.180
Ti+3_EDTA-4				
-1	---	3	C(10)H(12)N(2)O(8)Ti(1)	336.081

Ti+3_F-1(3)_(s) Titanium(III) fluoride; Titanium trifluoride				
0	---	1	F(3)Ti(1)	104.862
Ti+3_Gly-1				
2	---	2	C(2)H(4)N(1)O(2)Ti(1)	121.926
Ti+3_Gly-1_OH-1				
1	---	2	C(2)H(5)N(1)O(3)Ti(1)	138.934
Ti+3_H+1_EDTA-4				
0	---	2	C(10)H(13)N(2)O(8)Ti(1)	337.089
Ti+3_H+1_Gly-1				
3	---	2	C(2)H(5)N(1)O(2)Ti(1)	122.934
Ti+3_H+1_NTA-3				
1	---	1	C(6)H(7)N(1)O(6)Ti(1)	236.992
Ti+3_H+1(2)_EDTA-4				
1	---	2	C(10)H(14)N(2)O(8)Ti(1)	338.097
Ti+3_H+1(3)_Cl-1(6)				
0	---	1	Cl(6)H(3)Ti(1)	263.609
Ti+3_I-1(3)_(s) Titanium(III) iodide; Titanium triiodide				
0	---	1	I(3)Ti(1)	428.567
Ti+3_Leu-1				
2	---	1	C(6)H(12)N(1)O(2)Ti(1)	178.034
Ti+3_Malonic-2				
1	---	2	C(3)H(2)O(4)Ti(1)	149.914
Ti+3_Malonic-2(2)				
-1	---	2	C(6)H(4)O(8)Ti(1)	251.960

Ti+3_Malonic-2(3)				
-3	---	1	C(9)H(6)O(12)Ti(1)	354.006
Ti+3_MeMalon-2				
1	---	1	C(4)H(4)O(4)Ti(1)	163.940
Ti+3_OH-1				
2	---	2	H(1)O(1)Ti(1)	64.8743
Ti+3_OH-1_EDTA-4				
-2	---	3	C(10)H(13)N(2)O(9)Ti(1)	353.088
Ti+3_OH-1(2)				
1	---	1	H(2)O(2)Ti(1)	81.8817
Ti+3_OH-1(2)_EDTA-4				
-3	---	2	C(10)H(14)N(2)O(10)Ti(1)	370.096
Ti+3_OH-1(3)				
0	---	1	H(3)O(3)Ti(1)	98.8890
Ti+3_OH-1(3)_(s)				
0	---	2	H(3)O(3)Ti(1)	98.8890
Ti+3_Oxalic-2				
1	---	2	C(2)O(4)Ti(1)	135.887
Ti+3_Oxalic-2(2)				
-1	---	2	C(4)O(8)Ti(1)	223.906
Ti+3_Phenanth				
3	---	1	C(12)H(8)N(2)Ti(1)	228.076
Ti+3_Phenanth(2)				
3	---	1	C(24)H(16)N(4)Ti(1)	408.285

Ti+3_Phenanth(3)				
3	---	1	C(36)H(24)N(6)Ti(1)	588.494
Ti+3_Picolinic-1				
2	---	2	C(6)H(4)N(1)O(2)Ti(1)	169.970
Ti+3_Picolinic-1(2)				
1	---	3	C(12)H(8)N(2)O(4)Ti(1)	292.074
Ti+3_Picolinic-1(3)				
0	---	2	C(18)H(12)N(3)O(6)Ti(1)	414.177
Ti+3_Pro-1				
2	---	1	C(5)H(8)N(1)O(2)Ti(1)	161.991
Ti+3_SCN-1				
2	---	2	C(1)N(1)S(1)Ti(1)	105.945
Ti+3_SCN-1(2)				
1	---	2	C(2)N(2)S(2)Ti(1)	164.022
Ti+3_Ser-1				
2	---	1	C(3)H(6)N(1)O(3)Ti(1)	151.953
Ti+3_SO4-2				
1	---	2	O(4)S(1)Ti(1)	143.925
Ti+3_SO4-2(2)				
-1	---	2	O(8)S(2)Ti(1)	239.982
Ti+3_Val-1				
2	---	1	C(5)H(10)N(1)O(2)Ti(1)	164.007
Ti+3(2)_OH-1(2)				
4	---	2	H(2)O(2)Ti(2)	129.749

Ti+3(2)_S-2(3)_(s)				
0	---	1	S(3)Ti(2)	191.914
Ti+4 Titanium(IV) ion				
4	16043-45-1	111	Ti(1)	47.8670
Ti+4_2ThenTriFAcane-1				
3	---	2	C(8)H(4)F(3)O(2)S(1)Ti(1)	269.041
Ti+4_2ThenTriFAcane-1(2)				
2	---	2	C(16)H(8)F(6)O(4)S(2)Ti(1)	490.215
Ti+4_2ThenTriFAcane-1(3)				
1	---	2	C(24)H(12)F(9)O(6)S(3)Ti(1)	711.388
Ti+4_2ThenTriFAcane-1(4)				
0	---	1	C(32)H(16)F(12)O(8)S(4)Ti(1)	932.562
Ti+4_Br-1(4)_(g) Titanium(IV) bromide gas; Titanium tetrabromide gas				
0	---	1	Br(4)Ti(1)	367.483
Ti+4_Br-1(4)_(s) Titanium(IV) bromide; Titanium tetrabromide				
0	---	2	Br(4)Ti(1)	367.483
Ti+4_Cat-2(2)				
0	---	1	C(12)H(8)O(4)Ti(1)	264.060
Ti+4_Cat-2(2)_Oxalic-2				
-2	---	1	C(14)H(8)O(8)Ti(1)	352.080
Ti+4_Cat-2(3)				
-2	---	1	C(18)H(12)O(6)Ti(1)	372.157
Ti+4_CDTA-4				

0	---	1	C(14)H(18)N(2)O(8)Ti(1)	390.173
Ti+4_Cl-1				
3	---	1	Cl(1)Ti(1)	83.3200
Ti+4_Cl-1(2)_OH-1(2)				
0	---	1	Cl(2)H(2)O(2)Ti(1)	152.788
Ti+4_Cl-1(4)_(g) Titanium tetrachloride gas; Titanium(IV) chloride gas				
0	---	1	Cl(4)Ti(1)	189.679
Ti+4_Cl-1(4)_(l) Titanium(IV) chloride liquid				
0	---	1	Cl(4)Ti(1)	189.679
Ti+4_Cl-1(4)_(s) Titanium(IV) chloride; Titanium tetrachloride				
0	---	2	Cl(4)Ti(1)	189.679
Ti+4_Cupferron-1				
3	---	2	C(6)H(5)N(2)O(2)Ti(1)	184.985
Ti+4_Cupferron-1(2)				
2	---	2	C(12)H(10)N(4)O(4)Ti(1)	322.103
Ti+4_Cupferron-1(3)				
1	---	2	C(18)H(15)N(6)O(6)Ti(1)	459.221
Ti+4_Cupferron-1(4)				
0	---	1	C(24)H(20)N(8)O(8)Ti(1)	596.339
Ti+4_DTPA-5				
-1	---	1	C(14)H(18)N(3)O(10)Ti(1)	436.178
Ti+4_EDTA-4				
0	---	2	C(10)H(12)N(2)O(8)Ti(1)	336.081

Ti+4_EtDiAm_Pr2O1(4)				
4	---	2	C(14)H(40)N(2)O(4)Ti(1)	348.350
Ti+4_F-1_OH-1(2)				
1	---	3	F(1)H(2)O(2)Ti(1)	100.880
Ti+4_F-1(2)_OH-1(2)				
0	---	2	F(2)H(2)O(2)Ti(1)	119.879
Ti+4_F-1(3)_OH-1(2)				
-1	---	1	F(3)H(2)O(2)Ti(1)	138.877
Ti+4_F-1(4)				
0	---	1	F(4)Ti(1)	123.861
Ti+4_F-1(4)_(s) Titanium(IV) fluoride; Titanium tetrafluoride				
0	---	2	F(4)Ti(1)	123.861
Ti+4_F-1(4)_OH-1				
-1	---	1	F(4)H(1)O(1)Ti(1)	140.868
Ti+4_F-1(5)				
-1	---	1	F(5)Ti(1)	142.859
Ti+4_F-1(6)				
-2	---	4	F(6)Ti(1)	161.857
Ti+4_H+1_DTPA-5				
0	---	1	C(14)H(19)N(3)O(10)Ti(1)	437.186
Ti+4_H+1_F-1(6)				
-1	---	1	F(6)H(1)Ti(1)	162.865
Ti+4_H+1_OH-1(2)				
3	---	1	H(3)O(2)Ti(1)	82.8896

Ti+4_H+1_OH-1(2)_Citric-3(2)				
-3	---	2	C(12)H(13)O(16)Ti(1)	461.093
Ti+4_H+1_OH-1(2)_SO4-2				
1	---	1	H(3)O(6)S(1)Ti(1)	178.947
Ti+4_H+1(2)_Aniline(2)_Cat-2(3)				
0	---	1	C(30)H(28)N(2)O(6)Ti(1)	560.429
Ti+4_H+1(2)_OH-1(2)_Chromotropic-4(2)				
-4	---	1	C(20)H(12)O(18)S(4)Ti(1)	716.412
Ti+4_H+1(2)_OH-1(2)_SO4-2(2)				
0	---	1	H(4)O(10)S(2)Ti(1)	276.013
Ti+4_H+1(2)_OH-1(3)_234THB-4				
-1	---	1	C(7)H(7)O(8)Ti(1)	266.995
Ti+4_H+1(2)_OH-1(3)_234THB-4(2)				
-5	---	1	C(14)H(9)O(13)Ti(1)	433.085
Ti+4_H+1(2)_PAR-2				
4	---	2	C(11)H(9)N(3)O(2)Ti(1)	263.078
Ti+4_H+1(3)_OH-1_Chromotropic-4(3)				
-6	---	1	C(30)H(16)O(25)S(6)Ti(1)	1016.67
Ti+4_H+1(4)_OH-1(3)_24DHB-3(2)				
-1	---	1	C(14)H(13)O(11)Ti(1)	405.118
Ti+4_H+1(4)_PAR-2_Salicylic-2				
4	---	1	C(18)H(15)N(3)O(5)Ti(1)	401.201
Ti+4_H+1(4)_PAR-2_SulfSal-3				
3	---	1	C(18)H(14)N(3)O(8)S(1)Ti(1)	480.252

Ti+4_I-1(4)_ (s)				
Titanium(IV) iodide; Titanium tetraiodide				
0	---	2	I (4) Ti (1)	555.467
Ti+4_Lactic-1(2)_OH-1(3)				
-1	---	1	C (6) H (13) O (9) Ti (1)	277.031
Ti+4_O2-2				
2	---	2	O (2) Ti (1)	79.8658
Ti+4_OH-1				
3	---	8	H (1) O (1) Ti (1)	64.8743
Ti+4_OH-1_234TriOHPentDioic-2				
1	---	1	C (5) H (7) O (8) Ti (1)	242.973
Ti+4_OH-1_Malic-2				
1	---	1	C (4) H (5) O (6) Ti (1)	196.947
Ti+4_OH-1_SO4-2				
1	---	1	H (1) O (5) S (1) Ti (1)	160.932
Ti+4_OH-1_SulfSal-3				
0	---	2	C (7) H (4) O (7) S (1) Ti (1)	280.032
Ti+4_OH-1_SulfSal-3(2)				
-3	---	2	C (14) H (7) O (13) S (2) Ti (1)	495.189
Ti+4_OH-1(2)				
2	---	25	H (2) O (2) Ti (1)	81.8817
Ti+4_OH-1(2)_Cat-2_Oxalic-2				
-2	---	1	C (8) H (6) O (8) Ti (1)	277.998
Ti+4_OH-1(2)_SO4-2				
0	---	2	H (2) O (6) S (1) Ti (1)	177.939

Ti+4_OH-1(2)_SO4-2(2)				
-2	---	1	H(2)O(10)S(2)Ti(1)	273.997
Ti+4_OH-1(3)				
1	---	14	H(3)O(3)Ti(1)	98.8890
Ti+4_OH-1(3)_Malic-2				
-1	---	1	C(4)H(7)O(8)Ti(1)	230.962
Ti+4_OH-1(3)_PO4-3				
-2	---	1	H(3)O(7)P(1)Ti(1)	193.861
Ti+4_OH-1(4) Titanic(IV) acid				
0	---	9	H(4)O(4)Ti(1)	115.896
Ti+4_OH-1(4)_ (s) Titanium(IV) hydroxide; Titanium hydroxide				
0	20338-08-3	2	H(4)O(4)Ti(1)	115.896
Ti+4_OH-1(4)_PO4-3				
-3	---	1	H(4)O(8)P(1)Ti(1)	210.868
Ti+4_OH-1(5)				
-1	---	2	H(5)O(5)Ti(1)	132.904
Ti+4_Oxalic-2				
2	---	2	C(2)O(4)Ti(1)	135.887
Ti+4_Oxalic-2(2)				
0	---	2	C(4)O(8)Ti(1)	223.906
Ti+4_Oxine-1				
3	---	2	C(9)H(6)N(1)O(1)Ti(1)	192.020
Ti+4_Oxine-1(2)				
2	---	2	C(18)H(12)N(2)O(2)Ti(1)	336.173

Ti+4_Oxine-1(3)				
1	---	2	C(27)H(18)N(3)O(3)Ti(1)	480.325
Ti+4_Oxine-1(4)				
0	---	1	C(36)H(24)N(4)O(4)Ti(1)	624.478
Ti+4_PAR-2(2)				
0	---	2	C(22)H(14)N(6)O(4)Ti(1)	474.258
Ti+4_PAR-2(2)_Salicylic-2				
-2	---	1	C(29)H(18)N(6)O(7)Ti(1)	610.365
Ti+4_PAR-2(2)_SulfSal-3				
-3	---	1	C(29)H(17)N(6)O(10)S(1)Ti(1)	689.415
Ti+4_Pr2O1(4)				
4	---	2	C(12)H(32)O(4)Ti(1)	288.251
Ti+4_S-2(2)_ (s) Titanium(IV) sulfide; Titanium disulfide				
0	---	1	S(2)Ti(1)	111.987
Ti+4_Salicylaldoxime-2				
2	---	2	C(7)H(5)N(1)O(2)Ti(1)	182.989
Ti+4_Salicylaldoxime-2(2)				
0	---	1	C(14)H(10)N(2)O(4)Ti(1)	318.111
Ti+4_SCN-1				
3	---	1	C(1)N(1)S(1)Ti(1)	105.945
Ti+4_SulfSal-3(3)				
-5	---	1	C(21)H(9)O(18)S(3)Ti(1)	693.339
Ti+4(2)_EtDiAm_Pr2O1(8)				
8	---	1	C(26)H(72)N(2)O(8)Ti(2)	636.600

Ti (g) Titanium gas				
0	---	1	Ti (1)	47.8670
Ti (s) Titanium; Titanium, hexagonal; Titanium, alpha				
0	7440-32-6	80	Ti (1)	47.8670
Ti2O3 (aq. , s)				
0	---	1	O (3) Ti (2)	143.732
Ti2O3 (s) Dititanium trioxide				
0	---	5	O (3) Ti (2)	143.732
Ti2O5+2				
2	---	6	O (5) Ti (2)	175.731
Ti2O5+2_OH-1				
1	---	1	H (1) O (6) Ti (2)	192.738
Ti2O5+2_OH-1 (2)				
0	---	1	H (2) O (7) Ti (2)	209.746
Ti2O5+2_OH-1 (3)				
-1	---	1	H (3) O (8) Ti (2)	226.753
Ti2O5+2_OH-1 (4)				
-2	---	1	H (4) O (9) Ti (2)	243.760
Ti2O5+2_OH-1 (5)				
-3	---	1	H (5) O (10) Ti (2)	260.768
Ti2O5+2_OH-1 (6)				
-4	---	1	H (6) O (11) Ti (2)	277.775
Ti3O4+4				

4	---	1	O(4)Ti(3)	207.599
Ti3O5(s) Trititanium pentoxide, alpha				
0	---	2	O(5)Ti(3)	223.598
Ti3O5H3+5				
5	---	1	H(3)O(5)Ti(3)	226.622
Ti4O6H4+8				
8	---	1	H(4)O(6)Ti(4)	291.496
Ti4O7(s) Tetratitanium heptaoxide				
0	---	2	O(7)Ti(4)	303.464
TiB(s) Titanium monobromide				
0	---	1	B(1)Ti(1)	58.6770
TiB2(s) Titanium diboride				
0	---	1	B(2)Ti(1)	69.4870
TiC(s) Titanium monocarbide				
0	---	1	C(1)Ti(1)	59.8780
TiH(s)				
0	---	1	H(1)Ti(1)	48.8749
TiH2(s) Titanium dihydride; Titanium hydride				
0	---	3	H(2)Ti(1)	49.8829
TiN(s) Titanium mononitride				
0	---	1	N(1)Ti(1)	61.8737
TiO+1				

1	---	1	O(1)Ti(1)	63.8664
TiO+2 Titanium oxide ion; Titanyl ion				
2	12192-25-5	60	O(1)Ti(1)	63.8664
TiO+2_Acetic-1				
1	---	1	C(2)H(3)O(3)Ti(1)	122.911
TiO+2_CDTA-4				
-2	---	1	C(14)H(18)N(2)O(9)Ti(1)	406.172
TiO+2_Cl-1				
1	---	2	Cl(1)O(1)Ti(1)	99.3194
TiO+2_Cl-1(2)				
0	---	2	Cl(2)O(1)Ti(1)	134.772
TiO+2_Cl-1(3)				
-1	---	2	Cl(3)O(1)Ti(1)	170.225
TiO+2_Cl-1(4)				
-2	---	1	Cl(4)O(1)Ti(1)	205.678
TiO+2_ClO4-1				
1	---	1	Cl(1)O(5)Ti(1)	163.317
TiO+2_ClO4-1(2)				
0	---	1	Cl(2)O(9)Ti(1)	262.768
TiO+2_EDTA-4				
-2	---	1	C(10)H(12)N(2)O(9)Ti(1)	352.080
TiO+2_F-1				
1	---	2	F(1)O(1)Ti(1)	82.8648
TiO+2_F-1(2)				

0	---	2	F(2)O(1)Ti(1)	101.863
TiO+2_F-1(3)				
-1	---	2	F(3)O(1)Ti(1)	120.862
TiO+2_F-1(4)				
-2	---	1	F(4)O(1)Ti(1)	139.860
TiO+2_F-1(5)				
-3	---	1	F(5)O(1)Ti(1)	158.858
TiO+2_F-1(6)				
-4	---	1	F(6)O(1)Ti(1)	177.857
TiO+2_Gluconic-1				
1	---	1	C(6)H(11)O(8)Ti(1)	259.016
TiO+2_H+1_Acetic-1_PAR-2				
0	---	1	C(13)H(11)N(3)O(5)Ti(1)	337.114
TiO+2_H+1_Cat-2				
1	---	1	C(6)H(5)O(3)Ti(1)	172.971
TiO+2_H+1_CDTA-4				
-1	---	1	C(14)H(19)N(2)O(9)Ti(1)	407.180
TiO+2_H+1_EDTA-4				
-1	---	2	C(10)H(13)N(2)O(9)Ti(1)	353.088
TiO+2_H+1_OH-1_HOEDTA-3				
-1	---	1	C(10)H(17)N(2)O(9)Ti(1)	357.120
TiO+2_H+1_Pyrogallol-3				
0	---	1	C(6)H(4)O(4)Ti(1)	187.962
TiO+2_H+1_SO4-2				
1	---	1	H(1)O(5)S(1)Ti(1)	160.932

TiO+2_H+1_Tartaric-2				
1	---	2	C(4)H(5)O(7)Ti(1)	212.947
TiO+2_H+1(2)_CDTA-4				
0	---	1	C(14)H(20)N(2)O(9)Ti(1)	408.188
TiO+2_H+1(2)_Citric-3				
1	---	3	C(6)H(7)O(8)Ti(1)	254.984
TiO+2_H+1(2)_SulfSal-3(2)				
-2	---	1	C(14)H(8)O(13)S(2)Ti(1)	496.197
TiO+2_H+1(2)_SulfSal-3(3)				
-5	---	1	C(21)H(11)O(19)S(3)Ti(1)	711.354
TiO+2_H+1(2)_Tartaric-2(2)				
0	---	1	C(8)H(10)O(13)Ti(1)	362.027
TiO+2_H+1(2)_Tiron-4(2)				
-4	---	1	C(12)H(6)O(17)S(4)Ti(1)	598.276
TiO+2_H+1(2)_Tiron-4(3)				
-8	---	1	C(18)H(8)O(25)S(6)Ti(1)	864.474
TiO+2_H+1(4)_24DHB-3(2)				
0	---	1	C(14)H(10)O(9)Ti(1)	370.095
TiO+2_H+1(4)_Citric-3(2)				
0	---	2	C(12)H(14)O(15)Ti(1)	446.101
TiO+2_H2O2				
2	---	1	H(2)O(3)Ti(1)	97.8811
TiO+2_HO2-1				
1	---	1	H(1)O(3)Ti(1)	96.8731

TiO+2_NTA-3				
-1	---	1	C(6)H(6)N(1)O(7)Ti(1)	251.983
TiO+2_OH-1				
1	---	1	H(1)O(2)Ti(1)	80.8737
TiO+2_OH-1_HOEDTA-3				
-2	---	1	C(10)H(16)N(2)O(9)Ti(1)	356.112
TiO+2_OH-1(2)_(s)				
0	---	2	H(2)O(3)Ti(1)	97.8811
TiO+2_Oxalic-2				
0	---	2	C(2)O(5)Ti(1)	151.886
TiO+2_Oxalic-2(2)				
-2	---	3	C(4)O(9)Ti(1)	239.906
TiO+2_Salicylic-2				
0	---	1	C(7)H(4)O(4)Ti(1)	199.973
TiO+2_Salicylic-2(2)				
-2	---	1	C(14)H(8)O(7)Ti(1)	336.080
TiO+2_SO4-2				
0	---	2	O(5)S(1)Ti(1)	159.924
TiO+2_SO4-2(2)				
-2	---	2	O(9)S(2)Ti(1)	255.982
TiO+2_SulfSal-3				
-1	---	1	C(7)H(3)O(7)S(1)Ti(1)	279.024
TiO+2_Tartaric-2(2)				
-2	---	1	C(8)H(8)O(13)Ti(1)	360.011

TiO₂(8)_OH-1(12)				
4	---	1	H(12)O(20)Ti(8)	715.019
TiO(s) Titanium monoxide				
0	---	2	O(1)Ti(1)	63.8664
TiO₂(Anatase,s) Anatase; Titanium dioxide, tetragonal anatase; Titanium dioxide, anatase				
0	---	4	O(2)Ti(1)	79.8658
TiO₂(aq.,s) Titanium(IV) oxide, aquated; Titanium(IV) oxide aquated solid				
0	13463-67-7	2	O(2)Ti(1)	79.8658
TiO₂(Rutile,s) Rutile; Titanium dioxide, tetragonal rutile; Titanium dioxide, rutile				
0	1317-80-2	5	O(2)Ti(1)	79.8658
TiO₂(s)				
0	---	4	O(2)Ti(1)	79.8658
TiO₃.2H₂O(s)				
0	---	1	H(4)O(5)Ti(1)	131.896
Tiron-4 4,5-Dihydroxy-1,3-benzenedisulphonate ion; 4,5-Dihydroxy-1,3-benzenedisulfonate ion; 1,2-Dihydroxybenzene-3,5-disulphonate ion; 4,5-Dihydroxybenzene-1,3-disulphonate ion; Disodium 3,5-pyrocatecholdisulfonate ion; PDS				
-4	77310-82-8	263	C(6)H(2)O(8)S(2)	266.197
Val-1 L-Valinate ion; Valinate ion; L-2-Amino-3-methylbutanoate ion				
-1	---	582	C(5)H(10)N(1)O(2)	116.140
Zn+2 Zinc(II) ion				
2	23713-49-7	4977	Zn(1)	65.3820

Zn(s) Zinc; Zinc, hexagonal				
0	7440-66-6	131	Zn(1)	65.3820
Zn₂TiO₄(s) Zinc titanate; Dizinc titanium tetraoxide, tetragonal; Zinc titanium spinel; Dizinc titanium tetraoxide				
0	---	2	O(4)Ti(1)Zn(2)	242.629